

Machine Learning with Scikit-Learn

EEC 3150

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- Classification

Intro to ML

What is ML:

Machine learning is a field of data science/artificial intelligence which involves the use of statistical **models** to enable computers to learn and predict patterns in data

- A model is a function which can accurately map input data to output data
 $f(\text{input}) = \text{output}$
- Using a dataset of inputs/outputs to learn from, the model can learn the patterns (i.e., find a function f) and then, when given new input data, predict its output

Example: create a model to predict real estate prices

- Input data: square footage, location, garage?, number of floors
- Output data: the price (a continuous value)

Example: predict whether a patient has cancer

- Input data: age, gender, BMI, family history, environmental factors
- Output data: does/doesn't have cancer (a discrete label)

Some ML terms:

- Dataset: a collection of inputs/outputs
- Samples: entries in the dataset
(rows in the dataset)
- Features: the input data (usually X , a 2D array (N_{samp}, N_{feat}))
(columns in the dataset)
- Labels: the output data (usually y , a 1D array (N_{samp}))
(sometimes called “target”)
(a special column in the dataset)
- Model: a function f which predicts labels based on features
- Training: the process by which the model “learns” the patterns in a dataset
- Testing: the process by which a model’s performance/accuracy is evaluated
- Loss function: a function which evaluates the amount of error in the model
(sometimes called error, must be minimized,
training is minimizing the loss function)

Types of ML:

- Supervised learning: a type of learning where the dataset has features AND labels and the goal is to predict the labels based on the features
 - Regression (label is continuous)
 - Classification (label is categorical/discrete)
- Unsupervised learning: a type of learning where the dataset ONLY has features and the goal is to assign a set of reasonable labels
 - Clustering
- Generative learning: a type of learning where the goal is to create more data like the features
 - AI chatbots
 - AI image generators

Python libraries for ML:

- Scikit-learn (we'll use this one)
<https://scikit-learn.org/stable/modules/classes.html>
- Tensorflow, Keras, PyTorch (neural networks)
- XGBoost (gradient boosting)

Supervised Learning: Regression

Linear regression:

Linear regression is the process of fitting a linear model to the data

Single regression: fitting one feature (x) to the label

$$f(x) = ax + b$$

Multiple regression: fitting several features (x, x') to the label

$$f(x, x') = ax + bx' + c$$

Goal: find the slopes/intercept which minimize the loss

Supervised Learning: Regression

Linear regression:

Loss: sum of squared errors/residuals (SSE):

$$\text{loss} = \sum_{i=0}^{N_{\text{samp}}-1} \left(f(x_i) - y_i \right)^2$$

$$\text{loss}(a, b) = \sum_{i=0}^{N_{\text{samp}}-1} \left((ax_i + b) - y_i \right)^2$$

Minimize (ordinary least squares - OLS):

$$\min_{a,b} \text{loss}(a, b)$$

Other loss functions:

- Mean-squared error ($\text{MSE} = \text{loss}/N_{\text{samp}}$)
- Root mean-squared error ($\text{RMSE} = \sqrt{\text{MSE}}$)

Model evaluation:

There are several metrics for evaluating the quality of a linear regression model:

- Loss
- R^2 (square of Pearson correlation coefficient)

Supervised Learning: Regression

Linear regression:

Regression (or curve fitting) is the process of creating a model which uses features to predict continuous labels

Some example data:

	Features	Labels
	Sq ft	Price
0	1,500	210,000
1	2,000	280,000
2	1,650	260,000
3	2,200	320,000

Let

- $N_{samp} = 4$ be the number of samples
- $N_{feat} = 1$ be the number of features
- $X_{i,j}$ be j -th feature of the i -th sample
Ex: $X_{2,0} = 1,650$
- y_i be the i -th sample's label
Ex: $y_0 = 210,000$
- $f(\cdot)$ be our model
- loss: sum of squared errors (SSE)

$$\text{loss} = \sum_{i=0}^{N_{samp}-1} \left(f(X_{i,0}) - y_i \right)^2$$

Goal: find the model f which minimizes the loss

Supervised Learning: Regression

Linear regression:

Regression (or curve fitting) is the process of creating a model which uses features to predict continuous labels

Goal: find the model f which minimizes the loss

Some example data:

	Features		Labels
	Sq ft	Floors	Price
0	1,500	1	210,000
1	2,000	2	280,000
2	1,650	1	260,000
3	2,200	3	320,000

Let

- $N_{samp} = 4$ be the number of samples
- $N_{feat} = 2$ be the number of features
- $X_{i,j}$ be j -th feature of the i -th sample
Ex: $X_{3,1} = 3$
- y_i be the i -th sample's label
Ex: $y_0 = 210,000$
- $f(\cdot, \cdot)$ be our model
- loss: sum of squared errors (SSE)

$$\text{loss} = \sum_{i=0}^{N_{samp}-1} \left(f(X_{i,0}, X_{i,1}) - y_i \right)^2$$

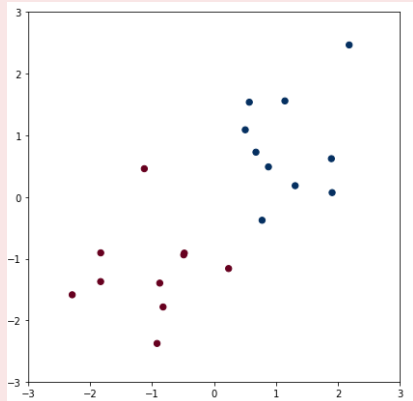
Supervised Learning: Classification

Supervised Learning: Classification

Classification:

Classification is the process of creating a model which uses features to predict discrete labels

Some example data:



What we're given:

- 2 features (on the x,y axes)
- A discrete label, 2 classes: red and blue

The goal:

- Create a model which can predict the class for new data

Some thoughts:

- The reds and blues each tend to cluster together
- New data that is firmly within the red cluster can be confidently classified as red (same for blue)
- New data which appears somewhere in b/t the two clusters is less clear how to classify

k Nearest Neighbors Classification (KNN):

Method:

- When given a new data point, find its k (an integer) nearest neighbors (the points closest to it)
- Assign the new point the label which is most frequent among its neighbors
- Choose the value of k which gives the lowest loss/highest accuracy

Loss function:

- There are many good loss functions for classification (cross entropy...)
- We will instead just maximize the accuracy (instead of minimizing loss)

Supervised Learning: Classification

Train/test splitting:

Motivation:

- We train our model using only the data we have at our disposal, but when your model actually gets used by someone, it will see NEW data
- We want our model to generalize to new data (it needs to do more than just predict the data it was trained on)
- To solve this, we split our data into two parts:
 - Training data: the data we use to create the model
 - Testing data: the data was use to evaluate the model's performance
- Once we find out what model parameters give the highest accuracy, we can then train on the whole dataset and let that be our final model

Some notes:

- Generally, we use an 80/20 split between training/test, respectively
- Generally, the training accuracy will be higher than the testing accuracy (makes sense since the model was designed with that data)

Accuracy vs. precision/recall:

- Sometimes accuracy is too simple of an evaluation metric
- We can also use precision and recall
- <https://towardsdatascience.com/accuracy-precision-recall-or-f1-331fb37c5cb9>